

SplitStack Project

Risk Assessment & Matrix

Project:	SplitStack MVP Development
Assessment Date:	February 2026
Project Duration:	6 Weeks (3 Sprints)
Version:	1.0
Status:	Confidential

Executive Summary

This risk assessment identifies 24 key risks across technical, operational, market, and strategic categories for the SplitStack MVP development project. Each risk has been evaluated based on probability of occurrence and potential impact, then plotted on a color-coded risk matrix.

The assessment reveals 5 **HIGH** risks, 11 **MEDIUM** risks, and 8 **LOW** risks. Critical areas requiring immediate attention include timeline management, third-party integration dependencies, and data security compliance.

Risk Level	Count	Percentage	Priority Action
HIGH	5	21%	Immediate mitigation required
MEDIUM	11	46%	Active monitoring & planning
LOW	8	33%	Standard management

Risk Matrix - Visual Overview

Risk Matrix Legend

Color	Risk Level	Action Required
HIGH	High Probability & High Impact	Immediate action & continuous monitoring
MEDIUM	Medium Probability/Impact	Develop mitigation plans
LOW	Low Probability & Low Impact	Monitor periodically

5×5 Risk Matrix

IMPACT → PROBABILITY ↓	Very Low (1)	Low (2)	Medium (3)	High (4)	Very High (5)
Very High (5)			R1, R4	R2, R3	
High (4)		R13	R5, R6, R7	R8	
Medium (3)	R20	R14, R15, R21	R9, R10, R11		
Low (2)	R22, R23	R16, R17, R24	R12		
Very Low (1)	R18, R19				

Note: Risk IDs (R1-R24) correspond to the detailed risk register on the following pages.

Detailed Risk Register

HIGH Risks

R1: Third-Party Integration Failures

Score: 15 (5×3)

Category: Technical

Description: Critical dependencies on Stripe, Auth0, AWS services may experience outages, API changes, or integration issues during development.

Mitigation Strategies:

- Test all integrations in Sprint 1
- Implement fallback authentication methods
- Use sandbox/test environments extensively
- Build abstraction layers for easy provider switching
- Allocate 20% buffer time for integration debugging

Risk Owner: CTO

R2: Timeline Slippage - Critical Features

Score: 20 (5×4)

Category: Schedule

Description: Core features (expense splitting, balance calculations) take longer than estimated, pushing back launch date.

Mitigation Strategies:

- Daily standups to identify blockers early
- Weekly sprint reviews with go/no-go decisions
- Pre-identify features to cut if timeline slips
- Implement strict scope control - defer non-MVP features
- Consider adding 1 additional developer if falling behind in Sprint 2

Risk Owner: Project Manager

R3: Data Security Breach or Vulnerability

Score: 20 (5×4)

Category: Technical

Description: Security vulnerability in authentication, payment processing, or data storage exposes user financial information.

Mitigation Strategies:

- Implement encryption at rest and in transit from day 1
- Use industry-standard auth providers (Auth0)
- Never store payment card data (use Stripe tokens)
- Conduct security code review before each sprint demo
- Perform penetration testing in Sprint 3
- Implement SQL injection and XSS prevention
- Set up automated security scanning in CI/CD

Risk Owner: CTO

R4: Key Team Member Unavailability

Score: 15 (5×3)

Category: Operational

Description: Critical team member (CTO, lead developer) becomes unavailable due to illness, emergency, or departure.

Mitigation Strategies:

- Cross-train team members on critical components
- Maintain comprehensive code documentation
- Use pair programming for complex features
- Keep detailed sprint notes and decision logs
- Identify backup resources who can step in
- Ensure no single point of knowledge failure

Risk Owner: CEO

R8: Database Performance Issues at Scale

Score: 16 (4×4)

Category: Technical

Description: PostgreSQL performance degrades with multiple concurrent groups, complex balance calculations, or large expense histories.

Mitigation Strategies:

- Design database schema with proper indexes from start
- Implement caching layer (Redis) for frequently accessed data
- Use connection pooling
- Load test with simulated user data (1000+ groups)
- Monitor query performance and optimize slow queries
- Plan for horizontal scaling (read replicas) post-MVP

Risk Owner: Backend Developer

MEDIUM Risks

R5: Mobile App Performance Issues

Score: 12 (4×3)

Category: Technical

Description: React Native app experiences lag, crashes, or poor performance on older devices or Android fragmentation.

Mitigation Strategies:

- Test on minimum supported devices (iPhone 8, Android 8)
- Implement lazy loading and code splitting
- Optimize image sizes and API payloads
- Use React Native performance profiling tools
- Set performance budgets (app load < 2 sec)

Risk Owner: Mobile Developer

R6: API Rate Limiting or Downtime

Score: 12 (4×3)

Category: Technical

Description: Backend APIs become unavailable or slow during peak usage, affecting user experience.

Mitigation Strategies:

- Implement rate limiting to prevent abuse
- Set up auto-scaling for API servers
- Use CloudWatch alarms for high latency/errors
- Implement graceful degradation for non-critical features
- Cache frequently accessed data
- Plan for 99.5% uptime SLA

Risk Owner: Backend Developer

R7: GDPR/CCPA Non-Compliance

Score: 12 (4×3)

Category: Compliance

Description: App violates data privacy regulations due to inadequate consent mechanisms, data retention, or user rights.

Mitigation Strategies:

- Implement explicit consent flows for data collection
- Build data export and deletion features
- Create privacy policy and terms of service
- Only collect essential data
- Consult with legal counsel on compliance
- Add cookie/tracking consent for web

Risk Owner: CEO / Legal

R9: Low User Adoption During Beta

Score: 9 (3×3)

Category: Market

Description: Difficulty recruiting beta testers or poor retention during beta phase indicates weak product-market fit.

Mitigation Strategies:

- Launch campus ambassador program in Sprint 1
- Incentivize beta testers (early access, swag, credits)
- Build referral mechanism into MVP
- Focus on college campuses with high co-living rates
- Collect feedback weekly and iterate quickly
- Set clear activation metrics (3+ expenses logged)

Risk Owner: Marketing Lead

R10: Competitive Product Launch

Score: 9 (3×3)

Category: Market

Description: Splitwise, Venmo, or new competitor launches similar features (AI, voting, gamification) during our development.

Mitigation Strategies:

- Move quickly - 6 week timeline minimizes risk window
- Focus on unique differentiators (AI + voting + gamification)
- Build strong community early (campus ambassadors)
- Emphasize superior UX and visual design
- Monitor competitor releases weekly

Risk Owner: CEO

R11: Budget Overrun - Infrastructure Costs

Score: 9 (3×3)

Category: Financial

Description: AWS, Stripe, or other service costs exceed budget due to higher-than-expected usage or pricing changes.

Mitigation Strategies:

- Set up billing alerts on all cloud services
- Use cost calculators to estimate monthly spend
- Optimize infrastructure (right-size instances)
- Use free tiers where possible during beta
- Monitor and cap API usage
- Reserve 25% budget buffer for overages

Risk Owner: CTO / CEO

R12: Inadequate Testing Coverage

Score: 6 (2×3)

Category: Operational

Description: Insufficient time allocated to testing results in bugs in production that damage user trust.

Mitigation Strategies:

- Allocate full week in Sprint 3 for testing
- Write unit tests alongside development (not after)
- Conduct daily manual testing during sprints

- Use automated E2E tests for critical flows
- Beta test with 50+ users to catch edge cases
- Set 90%+ test coverage target for core logic

Risk Owner: QA Engineer

R13: React Native Version Compatibility

Score: 8 (4×2)

Category: Technical

Description: Updates to React Native, Expo, or dependencies break compatibility during development.

Mitigation Strategies:

- Lock dependency versions in package.json
- Only update critical security patches
- Test updates in separate branch before merging
- Use stable versions (not latest/beta)
- Maintain changelog of dependency updates

Risk Owner: Mobile Developer

R14: Scope Creep - Feature Additions

Score: 6 (3×2)

Category: Operational

Description: Team or stakeholders push to add non-MVP features during development, delaying launch.

Mitigation Strategies:

- Maintain strict "MVP feature freeze" policy
- Document deferred features in backlog
- Require PM approval for any scope changes
- Remind team of 6-week hard deadline
- Show trade-offs: new feature = cut existing feature

Risk Owner: Project Manager

R15: App Store Rejection

Score: 6 (3×2)

Category: Market

Description: Apple or Google rejects app submission due to policy violations, content issues, or technical problems.

Mitigation Strategies:

- Review app store guidelines before development
- Ensure compliance with payment processing rules
- Submit early for TestFlight/Internal Testing
- Prepare clear app descriptions and privacy info
- Have legal review privacy policy
- Build in 1-week buffer for resubmission

Risk Owner: Mobile Developer

R21: Insufficient Documentation

Score: 6 (3×2)

Category: Operational

Description: Lack of technical or user documentation slows onboarding and support.

Mitigation Strategies:

- Write README and setup guides in Sprint 1
- Document APIs as they are built
- Create user guides in Sprint 3
- Use code comments for complex logic
- Record architecture decisions

Risk Owner: All Team

LOW Risks

R16: File Storage Limits Exceeded

Score: 4 (2×2)

Category: Technical

Description: S3 storage costs or limits exceeded due to receipt photo uploads.

Mitigation Strategies:

- Implement image compression before upload
- Set file size limits (5MB per receipt)
- Use lifecycle policies to archive old files
- Monitor storage usage weekly

Risk Owner: Backend Developer

R17: Push Notification Delivery Failures

Score: 4 (2×2)

Category: Technical

Description: Firebase Cloud Messaging fails to deliver notifications to some users.

Mitigation Strategies:

- Implement retry logic for failed notifications
- Use in-app notification center as backup
- Test across iOS and Android devices
- Monitor delivery rates

Risk Owner: Mobile Developer

R18: Seasonal Demand Fluctuation

Score: 1 (1×1)

Category: Market

Description: User acquisition varies by academic calendar (low during summer break).

Mitigation Strategies:

- Launch before summer semester starts
- Target year-round segments (young professionals)
- Plan marketing around back-to-school season
- Build referral mechanics for organic growth

Risk Owner: Marketing Lead

R19: Payment Processing Fees Higher Than Expected

Score: 1 (1×1)

Category: Financial

Description: Stripe fees or transaction costs reduce profit margins.

Mitigation Strategies:

- Review Stripe pricing structure upfront
- Pass 0.5% fee to users for settlements
- Optimize for ACH over card payments (lower fees)
- Negotiate enterprise rates at scale

Risk Owner: CEO

R20: Team Communication Breakdown

Score: 3 (3×1)

Category: Operational

Description: Remote or distributed team experiences communication gaps causing delays.

Mitigation Strategies:

- Daily standups (same time, required)
- Use Slack for async communication
- Document decisions in shared wiki
- Weekly sprint reviews with full team
- Use project management tools (Jira/Linear)

Risk Owner: Project Manager

R22: Brand Awareness Too Low

Score: 2 (2×1)

Category: Market

Description: Target users do not know SplitStack exists.

Mitigation Strategies:

- Focus on word-of-mouth and referrals
- Leverage campus ambassadors for awareness
- Create viral social media content
- Encourage group invitations (built-in growth)
- Optimize app store presence (ASO)

Risk Owner: Marketing Lead

R23: Customer Support Overwhelmed

Score: 2 (2×1)

Category: Operational

Description: Support volume exceeds team capacity during beta.

Mitigation Strategies:

- Build comprehensive FAQ and help center
- Use in-app chat for quick questions
- Set realistic support SLA (24-48 hours)
- Triage by severity (critical vs. general)
- Consider support chatbot for common issues

Risk Owner: CEO

R24: Minor UI/UX Issues

Score: 4 (2×2)

Category: Technical

Description: Small usability issues or design inconsistencies noticed during beta.

Mitigation Strategies:

- Conduct usability testing in Sprint 2
- Collect beta user feedback on UX

- Prioritize fixes: critical > nice-to-have
- Iterate based on user complaints
- Use analytics to identify friction points

Risk Owner: Product Designer

Risk Management Plan

Monitoring & Review

Risk management is an ongoing process throughout the 6-week development cycle. The following monitoring cadence will be implemented:

Daily Standups: Quick identification of emerging risks and blockers

Weekly Sprint Reviews: Formal risk assessment and mitigation status updates

Mid-Sprint Check-ins: Technical risk review (integrations, performance, security)

End-of-Sprint Retrospectives: Evaluate risk mitigation effectiveness and lessons learned

Escalation Process

Level 1 - Team Member: Identifies risk and attempts immediate mitigation

Level 2 - Risk Owner: Reviews risk, implements mitigation plan, monitors progress

Level 3 - Project Manager: Coordinates cross-functional mitigation, reallocates resources

Level 4 - CEO/Executive: Makes go/no-go decisions, approves scope changes or timeline adjustments

Contingency Budget & Time

To manage identified risks, the following contingencies have been allocated:

Time Buffer: 20% additional time allocated in Sprint 3 for critical bug fixes and rework

Budget Reserve: 25% financial buffer for infrastructure overruns and unexpected costs

Resource Flexibility: Pre-identified contractors available for emergency staffing

Scope Reserve: Pre-defined features that can be deferred if timeline slips (AI chat, OCR, advanced gamification)